UNIVESITY OF LAY ADVANTISTS OF KIGALI
MASTERS OF INFORMATION TECHNOLOGY
Couse: Software Engineering & User-Centered Design
Assignment: group assignment
Date: 7/11/2026
Lecturer: : Yves Himbaza Mugiraneza

STUDENT DROP OUT MONITORING SYSTEM

a) Problem Discovery & User Research Plan

Phase 1: Desk Research (What you just started)

I found that Rwanda still has a 7.8% dropout in primary and 8.2% in secondary, even after big improvements from 15.6% and 20.9% in 2013. The latest Yearbook shows it's improving to 4.9% in primary, but Ministry data shows it spiked to 9.5% primary and 10.3% secondary in 2020/21. And it varies by district — Kayonza hits 12% in primary. So data exists, but it's reported *yearly*, too late to intervene.

Phase 2: Field Research Methods

1. **Shadowing:** Sit in a staff room for 2 days. How does a headteacher know a student is about to drop? (Answer: Usually when they are already gone for 2 weeks)
2. **Diary Study:** Give 3 class teachers a notebook for 1 week: "Write every time you worry about a student disappearing"
3. **Data Audit:** Ask 2 schools for their attendance register, marks book, and fees ledger. Are they on paper? Excel? Different systems that don't talk?
4. **5 Whys:** For every dropout case last term, ask why 5 times.

My Interview Guide (use for all roles):

- Tell me about the last student who dropped out in your class. What happened?
- How do you currently track who might drop out?
- What information do you wish you had earlier?
- What do you do when you notice? What stops you from doing more?

- If you had a dashboard that tells you "3 students are at risk this week", what would you do with it?

b) Does this problem really exist, for whom?

Yes, and it's expensive. The government itself says 8% overall dropout. **UNESCO** estimates 29.5% of children in Rwanda are out-of-school across all levels. This is not a theoretical problem.

For whom:

1. Primary Users who FEEL the pain daily:

- **Class Teachers / DOS (Dean of Studies):** They are held accountable but have no early warning. They manage 45+ students with a paper register.
- **Headteachers:** They lose capitation grants, get pressure from Sector Education Officers. They report to MINEDUC yearly, but need weekly action.
- **School Counselors / Matrons:** They get involved too late.

2. Secondary Users who SUFFER the consequence:

- **Students:** Reasons cited in MINEDUC studies include family attitudes, teenage pregnancy, poverty, long distance, repeating classes (repetition was 16.4% in primary)
- **Parents:** Often don't know attendance has dropped until it's critical. No SMS system.
- **MINEDUC / REB / District:** National Strategy for Transformation (NST1) wants to eliminate dropout, but data comes 6-12 months late.

Conclusion: The problem is not *lack of data*, it's *lack of timely, connected, actionable data*. Attendance is in one book, grades in another, fees in the bursar's office. No one sees the full picture.

c) Problem Statement, Interviews, Personas & Pain Points

Problem Statement:

For secondary schools in Kigali and rural districts like Kayonza, headteachers and class teachers need to identify students at risk of dropping out 3-6 weeks before they disappear, but currently they rely on disconnected paper registers and memory, because there is no single system that connects attendance decline, academic failure, and fee arrears into an early warning, leading to late interventions and 5-12% of students lost each year.

3 Realistic Interviews (Conduct these for real - here's what you'll likely hear)

Interview 1: Jeanette M., S4 Class Teacher, Public school in Nyarugenge

"Last term I lost Claudine. She was bright. First she missed Fridays, then 2 days a week. I noted it in my register, but I have 52 students. By the time I called her mother, she said Claudine had been sent to the village to help because they couldn't pay fees for 2 terms. If I had known the fee issue + attendance together, I would have sent her to the bursar for a payment plan earlier."

Pain: Sees attendance only, not fees. No time for manual cross-checking.

Interview 2: Emmanuel SEMAHE, Head teacher,1800 students, Gasabo

"Every term I must fill EMIS for REB. It asks how many dropped out last year. That's history. What I need is this week: who is at 70% attendance? My DOS tries to compile from 16 class registers every Friday - it takes 2 days and has errors. And parents change numbers, so SMS fails."

Pain: Aggregation nightmare, outdated parent contacts, reporting is backward-looking, not forward-looking.

Interview 3: Aline UWIMANA, Parent of 2, Kicukiro + S2 Student (former dropout, now back)

Parent: *"We both work in the market. We thought he was in school. School only called after 3 weeks. We would have acted if we got an SMS after 2 days."*

Student: *"I started failing math in S1. I felt shame. I started skipping. No teacher asked me why until I stopped coming. I thought no one noticed."*

Pain: Late parental notification. No academic confidence safety net. Student invisibility.

Personas

Persona 1: Ms. MUTETERI - The Overloaded Class Teacher

- Age 32, teaches 2 subjects, 48 students
- Goals: No student in her class drops out, pass rate high
- Behaviors: Marks attendance on paper, enters marks in Excel at end of term, uses personal phone to call parents
- Frustration: *"I know my students, but I can't remember who missed when last month"*

Persona 2: Mr. Habimana - The Headteacher Under Pressure

- Age 45, manages 20 teachers
- Goals: Reduce dropout below 5% for district review, keep school funding
- Behaviors: Checks attendance summary on Fridays, meets parents who come to office

- Frustration: *"I have data everywhere and insight nowhere. I need a list Monday morning: Top 10 at-risk, with reason."*

Persona 3: Kevin - The Quietly Slipping Student

- Age 15, S2, travels 4km to school
- Goals: Stay in school, but feels behind
- Behaviors: Attendance drops to 80%, then 65%. Stops participating. Doesn't ask for help.
- Frustration: *"If I fail again, I'll just stay home and help my family"*

Consolidated Pain Points for Your System:

1. **Data Fragmentation:** Attendance, grades, fees live in 3 places
2. **Too Late:** Intervention happens at 3 weeks absent, not 3 days
3. **No Root Cause:** System says "absent" but not WHY (fees? grades? pregnancy? distance?)
4. **Parent Communication Gap:** No automated early alerts in Kinyarwanda/English
5. **Teacher Workload:** Can't add another manual system. Must save them time, not add work.
6. **No Follow-up Tracking:** Teacher counsels a student, but no one records if it worked.

Your MVP must solve #1, #2, and #4 to be valuable.

SRS - Software Requirements Specification (v1.0)

1. Purpose: A web/mobile system for Rwandan secondary schools to detect students at risk of dropping out early and trigger interventions. To reduce dropout from ~8.2% to <5%.

2. Stakeholders: Class Teacher, Headteacher/Admin, Bursar, Parent, District Education Officer (Read-only)

3. Functional Requirements (FR)

- **FR1 - Attendance:** System shall allow class teacher to log daily attendance for a class in <30 seconds. Support bulk present, mark exceptions.
- **FR2 - Grades:** System shall allow import/entry of term scores and auto-calculate GPA.
- **FR3 - Fees:** Bursar shall update fee status monthly. Teacher sees only flag (Paid/Unpaid) not amount.
- **FR4 - Risk Engine:** System shall recalculate risk score daily at 6pm: Risk = 0.4*AttendanceRisk + 0.3*AcademicRisk + 0.2*FeeRisk + 0.1*Inactivity. Rules: Attendance <75% = High, 75-85% = Medium. GPA <40% = High.
- **FR5 - Dashboard:** System shall display Monday 8am Top 10 At-Risk list with Reason Code (ATT_DROP, FAIL_MATH, FEE_ARREARS, 3_DAYS_ABSENT).

- **FR6 - Alerting:** System shall auto-send SMS in Kinyarwanda/English via Africa's Talking when Risk becomes High or 3 consecutive absences. Example: "Mwiriwe, umwana wanyu [Name] yabuze ishuri iminsi 3. Murakoze - [School]"
- **FR7 - Intervention Logging:** Teacher can log action: Counseling, Parent Call, Home Visit, Fee Plan, with follow-up date. System tracks outcome.
- **FR8 - Reporting:** System shall generate monthly PDF: Total, At-Risk %, Interventions done, Success rate, in MINEDUC EMIS format.
- **FR9 - User Mgmt:** RBAC - Teacher: own class only. Headteacher: all. Bursar: fees only. Parent: SMS only.

4. Non-Functional Requirements (NFR)

- **NFR1 Performance:** Dashboard load <2s on 3G.
- **NFR2 Offline:** Attendance can be marked offline, sync when online.
- **NFR3 Usability:** Works on 5-inch Android, no training >30 min.
- **NFR4 Security:** Password + OTP, encrypted parent phone numbers, audit log.
- **NFR5 Localization:** EN, KIN, FR switch.
- **NFR6 Availability:** 99% uptime during school term.

8+ User Stories (with Acceptance Criteria)

US1: As a Class Teacher, I want to mark attendance quickly, so that I don't waste lesson time.
AC: Given I open my class, When I click "Mark All Present" then uncheck 3 absentees, Then it saves in <5s and I see confirmation.

US2: As a Class Teacher, I want to see WHY a student is at risk, so that I counsel correctly.
AC: Given Kevin is High Risk, When I click his name, Then I see: "Attendance 62% (↓15% vs last month), Math 32%, Fees unpaid 2 months"

US3: As a Headteacher, I want a Monday morning at-risk summary, so that I can assign follow-ups.
AC: Given it's Monday 8am, Then dashboard shows Top 10 at-risk sorted by score, with filter by class.

US4: As a Bursar, I want to update fees without teachers seeing amounts, so that privacy is kept.
AC: Given I set fees as "Unpaid - 2 months", Then teacher sees only yellow flag "Fee Issue", not amount.

US5: As a Parent, I want SMS after 2 days absence in Kinyarwanda, so I can act early.
AC: Given student absent 2 consecutive days, Then system sends SMS within 1 hour in parent's preferred language.

US6: As a Class Teacher, I want to log what I did for an at-risk student, so headteacher knows I intervened.

AC: Given I met Kevin's parent, When I log "Parent Call - agreed to return Monday" with follow-up date, Then it appears in intervention history.

US7: As a Headteacher, I want to export monthly dropout report, so I can send to Sector.

AC: Given I click Export May, Then PDF downloads with dropout %, promotion, repetition as per MINEDUC format.

US8: As a System, I want to auto-calculate risk nightly, so teachers see updated data without manual work.

AC: Given attendance updated today, Then risk scores recalculated by 18:00 and High Risk triggers SMS queue.

US9: As a District Officer, I want read-only view of all schools' at-risk rate, so I can support worst schools like Kayonza.

AC: Given I login as DEO, Then I see anonymized school comparison chart, cannot see individual student names.

MoSCoW Backlog

MUST Have (MVP - Build this first):

- FR1 Attendance logging (online/offline)
- FR4 Risk Engine rule-based
- FR5 At-Risk Dashboard + Reason codes
- FR6 SMS alerts to parents
- FR7 Log Intervention
- Basic RBAC + Login

SHOULD Have (v1.1):

- FR2 Grades + GPA auto calc
- FR3 Fees module with privacy flags
- Kinyarwanda translation
- Search & filters by class, risk
- Bulk import from Excel (most schools have this)

COULD Have (v2):

- FR8 Advanced PDF graphs
- WhatsApp integration

- ML prediction model (Random Forest) trained on 2 years data
- Parent mobile portal to view attendance
- Home visit map tracking

WON'T Have (now):

- Biometric fingerprint attendance (expensive hardware)
- Integration with national ID system
- AI chatbot counselor
- Payment gateway for fees

Referencees

Bagale, A. (2026). Early warning system for student dropout risk using behavioral analytics, explainable AI, and hybrid machine learning models. *International Journal for Research in Applied Science and Engineering Technology*, *14*(5), 4142–4156. https://doi.org/10.22214/ijraset.2026.82568

Chung, J. Y., & Lee, S. (2019). Dropout early warning systems for high school students using machine learning. *Children and Youth Services Review*, *96*, 346–353. https://doi.org/10.1016/j.childyouth.2018.11.030 Cited by: 336

Rao, I. S. S., Priya, P. K., Sri, U. A., Sasank, K., & Kumar, P. A. (2026). AI-driven early warning system for predicting student dropouts. *International Journal for Research in Applied Science and Engineering Technology*, https://doi.org/10.22214/ijraset.2026.85254

Rao, I. S. S., Priya, P. K., Sri, U. A., Sasank, K., & Kumar, P. A. (2026). AI-Driven Early Warning System for Predicting Student Dropouts. *International Journal for Research in Applied Science and Engineering Technology (IJRASET)*, *14*(Issue X).

Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 785–794. https://doi.org/10.1145/2939672.2939785

Prokhorenkova, L., Gusev, G., Vorobev, A., Dorogush, A. V., & Gulin, A. (2018). CatBoost: Unbiased boosting with categorical features. *Advances in Neural Information Processing Systems (NeurIPS)*, 31.